15. period = 0.04096 ms

$f_{CLK\,I/O} = 100$ MHz

$f_{OCOB} = 1/period$

$f_{OCOB} = \frac{f_{CLK\,I/O}}{N \cdot (TOP+1)}$ $\Rightarrow \frac{1}{period} = \frac{f_{CLK\,I/O}}{N \cdot (TOP+1)} \Rightarrow$

$\Rightarrow N \cdot (TOP+1) = f_{CLK\,I/O} \cdot period = 100 \cdot 10^{6} \cdot 0.04096 \cdot 10^{-3} \Rightarrow$

$\Rightarrow N \cdot (TOP+1) = 0.04096 \cdot 10^{5} = 4096$

Avem 3 variante:

1. N=64 și TOP+1=64
2. N=256 și TOP+1=16
3. N=1024 și TOP+1=4

Vom alege prima variantă

TOP=63
TOP=OCR0A $\Rightarrow$ OCR0A = 63 $\Rightarrow$ WGM = 111

Factor de umplere de 25% $\Rightarrow$ OCR0B = $\frac{TOP}{4} \simeq 16 \Rightarrow$ COM0B = 10

N=64 $\Rightarrow$ CS0 = 011

TCCR0A $\rightarrow$ ob00 10 00 11
COM0B WGM(1,0)

TCCR0B $\rightarrow$ ob0000 1 011
WGM02 CS0

Scanned with CamScanner

```
TCCR0A equ 0x13
TCCR0B equ 0x18
OCR0A  equ 0x16
OCR0B  equ 0x15

ldi R16, 0b00100011
out TCCR0A, R16
ldi R16, 0b00001011
out TCCR0B, R16
ldi R16, 63
out OCR0A, R16
ldi R16, 16
out OCR0B, R16
```

Scanned with CamScanner